

LexAudit

*AI-Powered Legal Contract Risk Analysis with Tamper-Proof
Weilchain Audit Trails*

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The Problem

Issue	Consequence
Hidden risky clauses buried in legal text	Critical risks missed
No verifiable record of “who approved what”	Finger-pointing when disputes arise
Audit logs can be altered or deleted	No proof of compliance
Manual review is slow and expensive	Bottlenecks, high costs

Without a blockchain-based audit layer, every review is just a claim, not a proof.

Lawyer’s email can be deleted → no proof of approval.

TECHNOLOGY STACK

Layer	Technology	Purpose
Agent Orchestration	LangGraph (Python)	6-node state machine, conditional routing
Legal Intelligence	Groq Llama 3.3 70B	Clause extraction, risk scoring, reason generation
Blockchain SDK	weil_ai + weil_wallet	On-chain audit, wallet signing, auth headers
On-Chain Applets	Rust + WIDL	Two MCP applets deployed on marauder pod
Backend API	FastAPI	Bridges UI → agent, health checks, CORS
Frontend	Next.js (TypeScript)	Upload, live feed, human gate, report view
Testing	Pytest	23/23 passing tests (unit + integration)
Deployment	Render (backend), Vercel (frontend)	Live demo ready

How It Works :

1. **Upload** contract text.
2. **AI extracts clauses** and scores each (HIGH/MED/LOW).
3. **Human-in-the-loop gate** pauses if HIGH risk is found.
4. **Every step** is signed and recorded on Weilchain.
5. **Final report** generated with on-chain proof.

Key Differentiator :

- **Real AI** (Groq Llama 3.3 70B) not keyword matching.
- **Immutable audit** via Weilchain.
- **62 on-chain events** generated per contract — every step signed

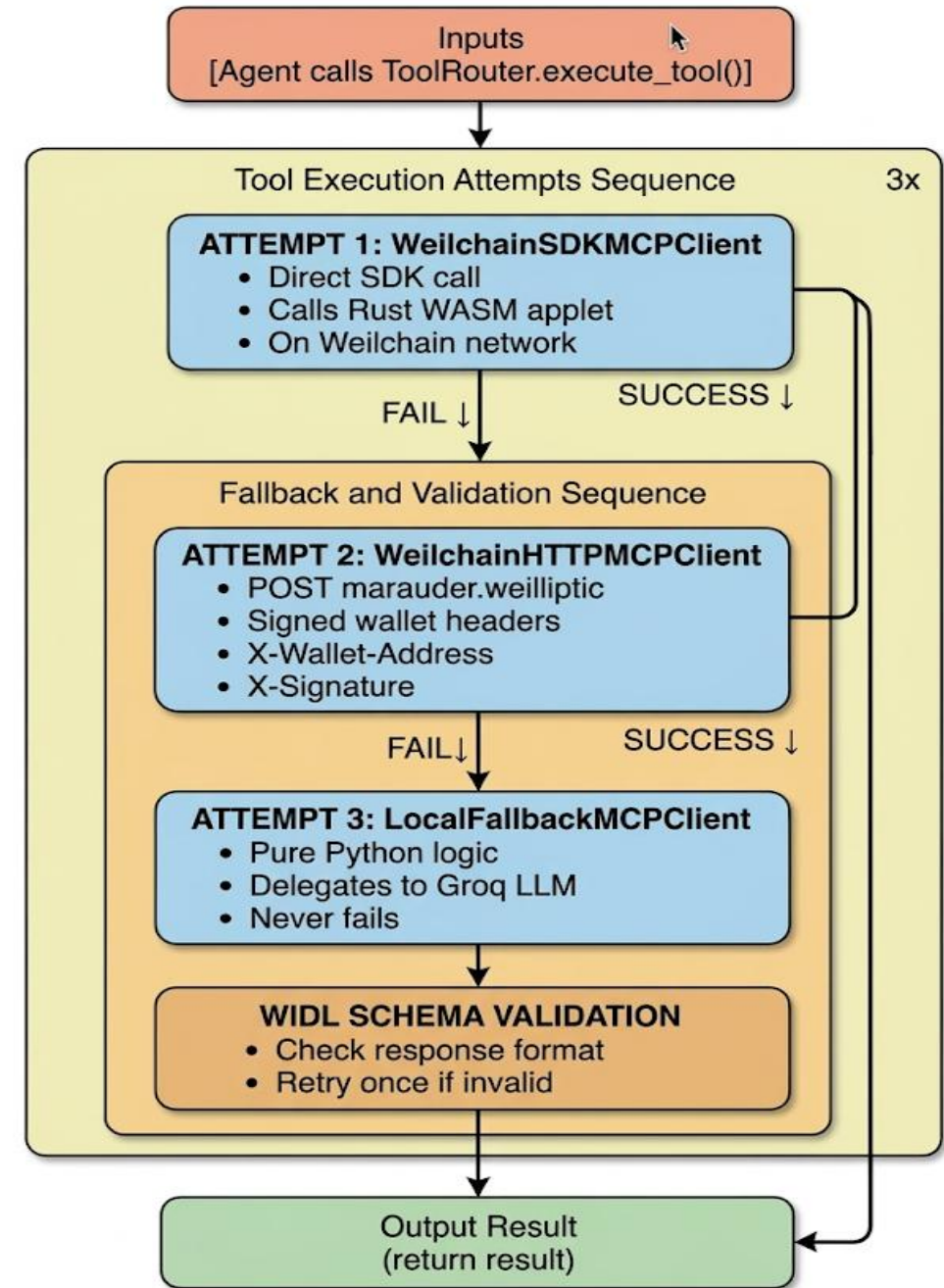


Figure 1: Tool Execution with nested fallbacks and schema validation.

Architecture

Layer 1: Agent Orchestration (LangGraph)

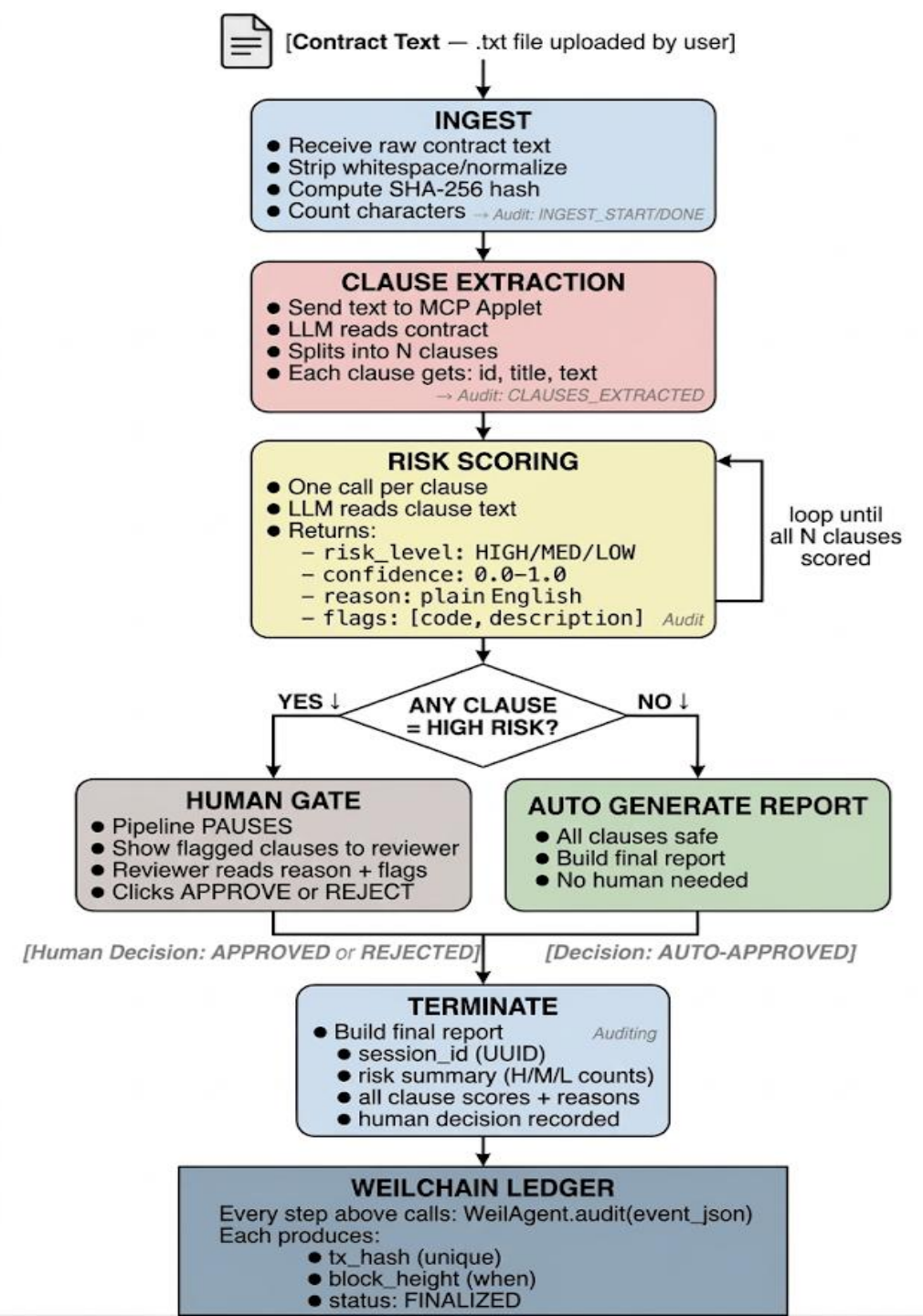
- 6-node state machine (INIT → INGEST → EXTRACT → RISK_LOOP → HUMAN_GATE → TERMINATE).
- Deterministic, step-budgeted, retry logic.

Layer 2: MCP Tool Router (weil_ai SDK)

- Attempt 1: WeilchainSDKMCP (Rust WASM applet)
- Attempt 2: WeilchainHTTPMCP (marauder pod)
- Attempt 3: LocalFallback + Groq (always works)
- All results validated against WIDL schemas.

Layer 3: On-Chain Audit Layer

- Every node → WeilAgent.audit(event).
- Wallet-bound identity (private_key.wc).
- Auth headers: X-Wallet-Address, X-Signature.
- Transactions visible on marauder.weilliptic.ai.



RESULTS & IMPACT

Metric	Value
Test Suite	23/23 passing 
On-Chain Events per Contract	62 (every step signed)
Clauses Extracted	Up to 23 per contract
Risk Detection	12 HIGH clauses in sample (all correctly flagged)
Human Gate	Triggers on ANY HIGH risk, pauses for review
Audit Trail	Wallet address explorer, tx hashes for every step
Sample Flagged Clauses	Pricing Authority → HIGH, IP Capture → HIGH, Non -Compete (10yr) → HIGH